UNIVERSITY OF WATERLOO

MTE 203 - Lecture 4

Midterm Exam Spring 2025, Problem 2

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a. Find an equation of the plane through the line of intersection of the planes

$$x - z = 1 \qquad \mathcal{P}_1$$
$$y + 2z = 3 \qquad \mathcal{P}_2$$

and perpendicular to the plane $x + y - 2z = 1$. $\mathcal{P}_3$

b. What is the distance between point $P(2,1,4)$ and this plane.

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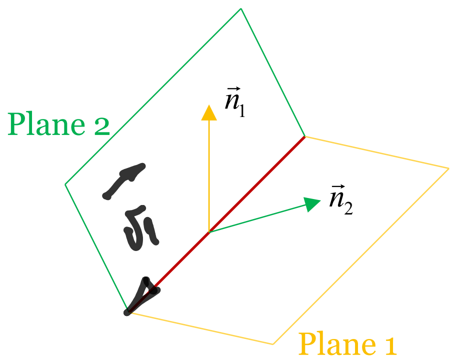
a. From Lecture 3, we learned that to determine the equation of a plane using vectors, we need 2 known vectors in that plane.

First known vector:



$$\vec{v}_1 = \vec{n}_1 \times \vec{n}_2$$

$$\vec{n}_1 \Big|_{P_1} = (1, 0, -1)$$

$$\vec{n}_2 \Big|_{P_2} = (0, 1, 2)$$

$$\vec{v}_1 = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 0 & -1 \\ 0 & 1 & 2 \end{vmatrix} = \hat{k} + \hat{i} - 2\hat{j}$$

$$\vec{v}_1 = (1, -2, 1)$$

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Second known vector:

PARALELL TO THE NORMAL TO Π_3

$$\vec{v}_2 : \vec{n}_3 = (1, 1, -2)$$

$$\vec{v}_2 = (1, 1, -2)$$

The normal to the plane we are looking for will therefore be given by,

$$\vec{n} = \vec{v}_1 \times \vec{v}_2$$

$$\vec{n} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 1 \\ 1 & 1 & -2 \end{vmatrix} = \cdots = (3, 3, 3)$$

OR : $\vec{n} = (1, 1, 1)$

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A POINT IN THE PLANE CAN BE FOUND FROM INTERSECTION OF $\mathcal{P}_1$ AND $\mathcal{P}_2$

$$\left.\begin{array}{l} x - z = 1 \\ y + 2z = 3 \end{array}\right\} \text{ IF } z = 0 \quad \left\{\begin{array}{l} x = 1 \\ y = 3 \end{array}\right.$$

$$P_0\ (1, 3, 0)$$

THE EQ' OF THE PLANE IS,

$$1(x-1) + 1(y-3) + 1(z) = 0$$

$$x + y + z = 4$$

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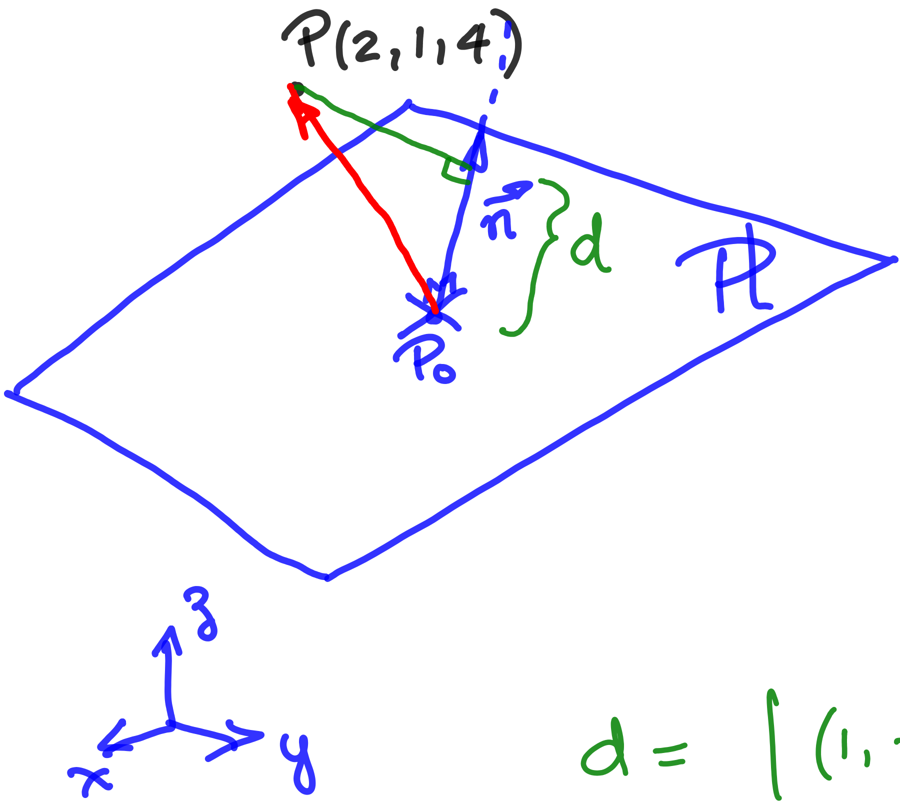
b. What is the distance between point $P(2,1,4)$ and this plane.



$$d = |\vec{P_0P} \cdot \hat{n}|$$

$$\vec{P_0P} = (2,1,4) - \underbrace{(1,3,0)}_{P_0}$$

$$\vec{P_0P} = (1,-2,4)$$

$$\hat{n} = \frac{(1,1,1)}{\sqrt{3}}$$

$$d = \left| (1,-2,4) \cdot \frac{(1,1,1)}{\sqrt{3}} \right| = \ldots \sqrt{3}$$

Thank You